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Assignment 2

Lookup Functions

Question 1

- A manager wants to review compensation (Salary) for the following employees
- Using Vlookup, return the following:

Employee No	Employee Name	Monthly Salary	Job Rate
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```
=VLOOKUP(A4,Employees,MATCH($B$3,Employees[#Headers],0),FALSE)
```

```
=VLOOKUP(A4,Employees,MATCH($C$3,Employees[#Headers],0),FALSE)
```

```
=VLOOKUP(A4,Employees,MATCH($D$3,Employees[#Headers],0),FALSE)
```

Q1:

A manager wants to review compensation (Salary) for the following employees

Using Vlookup, return the following:

Employee No	Employee Name	Monthly Salary	Job Rate
12	Muhamad Iad	\$2,016	1
64	Ahmad Alealbi	\$2,304	5
87	Muhamad Khayr	\$1,166	5
18	Farahad Husayn	\$1,295	3
33	Muhamad Eurul	\$1,551	1
2	Omar Hishan	\$3,247	1
24	Ahmad Ahmad	\$2,682	4.5

Lookup Functions

Question 2

- The HR department needs to identify where employees are located.
- Using Xlookup, retrieve the following information:

Employee No	Employee Name	Country	Center
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```
=XLOOKUP(F4,Employee_No,Employee_Name,"Not Found",0)
```

```
=XLOOKUP(F4,Employee_No,Country,"Not Found",0)
```

```
=XLOOKUP(F4,Employee_No,Center,"Not Found",0)
```

Q2:

The HR department needs to identify where employees are located.

Using Xlookup, retrieve the following information:

Employee No	Employee Name	Country	Center
55	Eubay Alttahir	Egypt	North
5	Ahmad Bikri	Egypt	Main
33	Muhamad Eurul	Egypt	North
11	Muhamad Alqadal	Syria	Main
8	Muhamad Alaya	Egypt	Main
104	Rami Abu Adhan	Syria	Main
49	Rim Almusaa	Egypt	West

Lookup Functions

Question 3

- Sometimes HR receives an employee's name instead of their employee number.
- Using Xlookup, find the following information:

First Name	Job Rate	Department	Annual Salary
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```
=XLOOKUP(K4,First_Name,Job_Rate,"Not Found",0)
```

```
=XLOOKUP(K4,First_Name,Department,"Not Found",0)
```

```
=XLOOKUP(K4,First_Name,Annual_Salary,"Not Found",0)
```

Q3:

Sometimes HR receives an employee's name instead of their employee number.

Using Xlookup, find the following information:

First Name	Job Rate	Department	Annual Salary
Dijwar	5	Product Development	\$8,928
Jawaher	5	Quality Assurance	\$40,968
Farahad	3	Quality Control	\$15,540
Layzaan	3	Facilities/Engineering	\$24,204
Almuthanaa	5	Account Management	\$30,876
Batual	1	Manufacturing	\$29,616
Bushraa	1	Facilities/Engineering	\$35,832

Lookup Functions

Question 4

- Operations management wants to identify employees working the most overtime.
- Using Vlookup, return the following:

Employee No	Department	Overtime Hours	Job Rate
-------------	------------	----------------	----------

```
=VLOOKUP(A15,Employees,MATCH($B$14,Employees[#Headers],0),FALSE)
```

```
=VLOOKUP(A15,Employees,MATCH($C$14,Employees[#Headers],0),FALSE)
```

```
=VLOOKUP(A15,Employees,MATCH($D$14,Employees[#Headers],0),FALSE)
```

Q4:

Operations management wants to identify employees working the most overtime.

Using Vlookup, return the following:

Employee No	Department	Overtime Hours	Job Rate
309	Quality Control	4	5
66	Manufacturing	3	4.5
6	Product Development	8	3
32	Environmental Compliance	48	3
22	Manufacturing Admin	109	3
3	Major Mfg Projects	192	2
24	Quality Assurance	32	4.5

Lookup Functions

Question 5

- The HR department wants to review how long certain employees have
- been working at the company.
- Use Index and Match to retrieve the following information:

Employee No	Employee Name	Start Date	Years at the Company
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=INDEX(Employees,MATCH(F15,Employee_No,0),MATCH(\$G\$14,Employees[#Headers],0))

=INDEX(Employees,MATCH(F15,Employee_No,0),MATCH(\$H\$14,Employees[#Headers],0))

=INDEX(Employees,MATCH(F15,Employee_No,0),MATCH(\$I\$14,Employees[#Headers],0))

Q5:

The HR department wants to review how long certain employees have been working at the company.

Use Index and Match to retrieve the following information:

Employee No	Employee Name	Start Date	Years at the Company
37	Rima AlAsfar	1/27/2016	10
1	Ghadir Hmshw	4/4/2018	8
34	Eizat Ghanim	6/21/2017	8
55	Eubay Alttahir	9/16/2016	9
45	Ahmad Laylana	11/27/2016	9
5	Ahmad Bikri	3/11/2020	6
67	Muhamad Jawish	11/25/2019	6

Lookup Functions

Question 6

- The HR manager wants to investigate employee attendance
- Use Index and Match to retrieve the following information:

Employee No	Employee Name	Sick Leaves	Unpaid Leaves
-------------	---------------	-------------	---------------

```
=INDEX(Employees,MATCH(K15,Employee_No,0),MATCH($L$14,Employees[#Headers],0))
```

```
=INDEX(Employees,MATCH(K15,Employee_No,0),MATCH($M$14,Employees[#Headers],0))
```

```
=INDEX(Employees,MATCH(K15,Employee_No,0),MATCH($N$14,Employees[#Headers],0))
```

Q6:

The HR manager wants to investigate employee attendance

Use Index and Match to retrieve the following information:

Employee No	Employee Name	Sick Leaves	Unpaid Leaves
5	Ahmad Bikri	0	5
35	Ahmad Swyd	1	0
267	Rannym Tyru	0	0
387	Lwy Abu	2	0
88	Samah Almaydani	0	0
100	Lina Aljabr	0	0
73	Shahad Shanan	0	2

If And Nested If

Question 1

- HR wants to identify employees who worked significant overtime.
- Create a column called Overtime Status.
- If Overtime Hours > 100 → "High Overtime"
- Otherwise → "Normal"

Overtime Status ▼

```
=IF(@Overtime_Hours>100,"High Overtime","Normal")
```


If And Nested If

Question 2

- Finance wants to classify employees based on Monthly Salary.
- Create a column called Salary Category.

Monthly Salary	Category
> 7000	High
4000–7000	Medium
< 4000	Low

Salary Category ▼

```
=IF(@Monthly_Salary>7000,"High",IF(AND(@Monthly_Salary>=4000,@Monthly_Salary<=7000),"Medium","Low"))
```


If And Nested If

Question 3

- HR wants to classify employee experience.
- Create a column called Experience Level.

Years	Level
≥ 10	Senior
5-9	Mid-Level
< 5	Junior

Experience Level ▼

```
=IF(@Years>=10,"Senior",IF(AND(@Years>=5,@Years<=9),"Mid-Level","Junior"))
```


If And Logical Functions

Question 1

- The company wants to identify employees who worked many overtime hours and have long experience.
- Create a column called Hardworking Employee.
- Rule: If Years ≥ 5 AND Overtime Hours > 50 , then "Yes"
- otherwise "No"

Hardworking Employee ▼

```
=IF(AND(@Years>=5,@Overtime_Hours>50),"Yes","No")
```


If And Logical Functions

Question 2

- HR wants to flag employees who have attendance concerns.
- Create a column called Attendance Warning.
- Rule: If Sick Leaves > 3 OR Unpaid Leaves > 3 then "Warning"
- otherwise "OK"

Attendance Warning ▼

```
=IF(OR(@Sick_Leaves>3,@Unpaid_Leaves>3),"Warning","OK")
```


If And Logical Functions

Question 3

- Finance wants to determine bonus eligibility.
- Create a column called Experience Level.
- Rule: If Years ≥ 5 AND Job Rate ≥ 3 AND (Overtime Hours > 50 OR Monthly Salary > 7000) then "Eligible"
- otherwise "Not Eligible"

Experience Level2 ▼

```
=IF(AND(@Years>=5,@Job_Rate>=3,OR(@Overtime_Hours>50,@Monthly_Salary>7000)),"Eligible","Not Eligible")
```